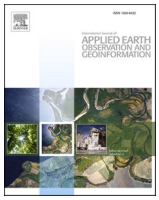
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Integrating earth observation data into the tri-environmental evaluation of the economic cost of natural disasters: a case study of 2025 LA wildfire

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ABSTRACT

Wildfires in urbanized regions, particularly within the wildland–urban interface, have significantly intensified in frequency and severity, driven by rapid urban expansion and climate change. This study aims to provide a comprehensive, fine-grained evaluation of the recent 2025 Los Angeles wildfire's impacts, through a multi-source, tri-environmental framework in the social, built and natural environmental dimensions. This study employed a spatiotemporal wildfire impact assessment method based on daily satellite fire detections from the Visible Infrared Imaging Radiometer Suite (VIIRS), tri-environmental data from OpenStreetMap, and high-resolution dasymetric population modeling to capture the dynamic progression of wildfire events in two distinct Los Angeles County regions, Eaton and Palisades, which occurred in January 2025. The modelling result estimated that the potential economic exposure reached approximately $48.55 billion, with the highest single-day exposure recorded on January 8 in both districts. Our modelling results highlight early, severe ecological and infrastructural damage in Palisades, as well as delayed, intense social and economic disruptions in Eaton. This tri-environmental framework underscores the necessity for tailored, equitable wildfire management strategies, enabling more effective emergency responses, targeted urban planning, and community resilience enhancement. Our study contributes a highly replicable tri-environmental framework for evaluating potential natural, built, and social environmental exposure to natural disasters, which can be applied to future risk profiling, hazard mitigation, and environmental management in the era of climate change.

1. Introduction

Wildfire has become an increasingly urban problem in the western United States as metropolitan growth pushes farther into flammable landscapes: in California alone, the wildland–urban interface (WUI) expanded by more than 30% between 1990 and 2020, situating millions of residents and billions of dollars in assets in places where severe fire is no longer an anomaly but an annual expectation (Radeloff et al., 2018; Chow et al., 2022). Recent events underscore the stakes: the 2018 Camp Fire leveled the town of Paradise; the 2021 Dixie Fire scorched nearly one million acres; and a series of fast-moving blazes in early 2025 threatened multiple neighborhoods in Los Angeles County (Matt, 2024; Tutella, 2021). Beyond the immediate loss of life and property, such fires interrupt economic activity, degrade air quality, and exacerbate long-term ecological stressors such as invasive species spread and post-fire soil erosion (Gonçalves et al., 2025). Yet most wildfire impact assessments tend to treat these consequences in isolation—mapping burn scars one season at a time, tabulating insurance claims, or profiling social vulnerability—without analyzing how the natural, built and social environment interact dynamically during the course of a single fire event. This research addresses this gap by contributing a fine-grained, tri-environmental framework that captures the full spatial and temporal complexity of the wildfire impact.

A substantial body of research has advanced our understanding of wildfire behavior and consequences, but often through discipline-specific lenses. Ecological studies focus on vegetation loss and land degradation using spectral indices such as differenced normalized burn ratio (dNBR) (Ibrahim et al., 2024), while infrastructure assessments examine damage to roads, buildings, and utilities. Social vulnerability research highlights the disparate capacities of populations to respond to

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and recover from fire events, often linked to age, income, ethnicity, and housing conditions (Moore et al., 2023). However, these studies typically operate at coarse spatial or temporal scales or rely on static data snapshots that fail to capture the evolving nature of fire exposure. A few recent researchers explored integrative approaches that combine environmental, built, and social variables particularly in the context of floods and heatwaves (Raymond et al., 2020; Wang et al., 2023a,b). In wildfire research, such comprehensive and real-time evaluations across environmental, built, and social dimensions are much needed to support timely response strategies for complex events such as the 2025 LA Wildfire.

Advances in earth-observation technologies and open geospatial data now offer an opportunity to close this integration gap. Near-real-time satellite sensors such as Moderate Resolution Imaging Spectroradiometer (MODIS) and Visible Infrared Imaging Radiometer Suite (VIIRS) provide frequent thermal detections that enable tracking of fire progression at daily intervals. Platforms such as the Fire Information for Resource Management System (FIRMS), OpenStreetMap (OSM), and the National Land Cover Database (NLCD) offer globally accessible layers on active fire points, built environment, and natural environment respectively (Haklay and Weber, 2008; Liu et al., 2024; NASA., 2025). Furthermore, dasymetric mapping techniques have made it possible to downscale census populations to grid resolutions as fine as 10–30 m, redistributing demographic data across heterogeneous land cover types and revealing who lives where with much greater precision than traditional census tracts (Swanwick et al., 2022; Depsky et al., 2022). However, despite the availability of these tools, few studies have developed workflows that simultaneously and dynamically incorporate these multi-source data streams to assess daily wildfire impacts across natural, built and social environments.

To address these knowledge gaps, this study introduces and implements a multi-source, tri-environmental framework for wildfire impact analysis, integrating natural, built, and social environmental evaluation at a fine spatial and temporal resolution (20 m by 20 m on the daily basis). The framework is demonstrated through application to two distinct WUI regions in Los Angeles County, Eaton and Palisades, each characterized by contrasting topography, vegetation, settlement patterns, and demographic profiles. The reconstructed daily wildfire perimeters, derived from VIIRS thermal anomalies and refined using CAL FIRE official boundaries, are overlaid with NLCD-based land cover data, OSM-derived built environment features (including road networks and building footprints), Foursquare Points of Interest (POI) data, and a 20-meter resolution dasymetric population grid. This layered spatial analysis enables the disaggregation of wildfire impacts by environment type and population vulnerability at a daily basis.

Through the tri-environmental evaluation, this paper makes three key contributions. First, it presents a reproducible framework for daily-scale fire boundary reconstruction and impact quantification using publicly available data. Second, it operationalizes a tri-environmental framework that captures the interdependencies among ecological exposure, infrastructure disruption, and social vulnerability, which are too often treated separately in disaster research. Third, it demonstrates the value of high-resolution, intra-urban comparative analysis through a case study of Eaton and Palisades, offering practical insights into how wildfire risk manifests differently across neighborhoods. Ultimately, the findings underscore the need for integrated, spatially explicit, and equity-aware wildfire management strategies that align emergency response, urban planning, and community resilience efforts.

2. Literature review

2.1. Wildfire impacts in Wildland–Urban interface (WUI) zones

Wildfires in the WUI, where urban structures intersect with wildland vegetation, are becoming increasingly devastating. In California, rapid WUI expansion into flammable landscapes has sharply elevated risk (Kumar et al., 2025). From 2000 to 2018, about 54,000 U.S. buildings were lost to wildfires, with 86–97% located in the WUI (Caggiano et al., 2020). California's WUI growth and climate change have together increased extreme fire conditions 4.1-fold since 1990 (Kumar et al., 2025). Most major fires now originate within 1 km of WUI areas, underscoring wildfire as a predominantly WUI-driven hazard (Radeloff et al., 2018).

Damage patterns in WUI fires are multifaceted. Structural losses often occur in clusters, destroying entire neighborhoods, as seen in the 2018 Camp Fire and 2025 Los Angeles fires. Beyond property loss, wildfires cause severe ecological degradation, including habitat fragmentation, biodiversity loss, erosion, and nutrient depletion (Troy et al., 2022), often worsened by human activities like fire suppression and invasive species. Socially vulnerable populations face disproportionate wildfire exposure, which rose significantly from 2000 to 2021, especially among low-income, elderly, and mobility-impaired residents in California (Modaresi Rad et al., 2023; Masri et al., 2021). However, most of the previous WUI studies focused on a single perspective of the built environment (home ignition risks, building materials, defensible space), natural environment (fire regimes, ecological impacts), or social environment (vulnerability mapping, evacuation logistics)—limiting the comprehensive understanding of wildfire impacts within WUI zones.

2.2. Advances in daily wildfire monitoring and mapping

Recent technological advances facilitate daily-scale wildfire monitoring, improving management and research capabilities. Historically, burned areas were assessed post-event, but current satellite technologies such as NASA's MODIS and the VIIRS now offer near real-time data at resolutions of approximately 1 km and 375 m, respectively. NASA's FIRMS also provides continuous global fire hotspot data, significantly enhancing wildfire situational awareness (NASA FIRMS, 2025).

Advanced techniques now convert satellite fire detections into actionable insights. Algorithms cluster thermal pixels into dynamic perimeters, enabling real-time updates—some systems generate fire boundaries every 12 h, improving fire behavior monitoring (Chen et al., 2022). This approach offers faster insights than post-event imagery, aiding daily fire tracking and model evaluation. Spatial tools such as Kernel Density Estimation (KDE) reveal ignition hotspots and shifting fire patterns, both historically and in real time. Real-time KDE supports emergency response by highlighting active fire zones. Integrating fire perimeters with social and built environment data via tools Ready-Mapper, helps managers identify threats to vulnerable populations and critical services, enabling timely, targeted evacuation and resource deployment (Schroeder et al., 2023). Beyond remote sensing, complementary operational approaches strengthen the daily monitoring-to-management pipeline. Tampekis et al. (2023) propose a performance-based wildfire engineering framework that can inform built-environment resilience and infrastructure planning in fire-prone urban regions. Sakellariou et al. (2017) optimize the placement of fire watchtowers for early detection, highlighting the value of hybrid remote–local monitoring to complement satellite detections. Sakellariou et al. (2023) model the optimal deployment of firefighting resources and show how topography shapes initial-attack effectiveness, linking daily detection capabilities to terrain-informed suppression strategies.

However, despite available technology, literature gaps persist. Most wildfire studies remain retrospective, lacking full utilization of daily satellite data granularity. There is limited published research operationalizing daily tracking in impact assessments or systematically integrating fire behavior data with socio-demographic information. Bridging daily satellite monitoring with performance-based design, optimized watchtower siting, and resource-deployment models could substantially enhance end-to-end wildfire assessment and response.

2.3. Tri-Environmental framework for disaster research

The tri-environmental framework, encompassing the natural, built, and social, offers a holistic approach to disaster analysis, recognizing disasters as products of interconnected systems rather than isolated phenomena (Wang et al., 2023). Rooted in social–ecological theory (Stokols, 1996), the framework is increasingly applied in disaster research as a way to link these three dimensions into a unified assessment.

Recent studies applying the tri-environmental framework illustrate its effectiveness across hazards. For instance, Wang et al. (2023) apply the framework to Australian heatwave risk assessments, integrating hazard metrics (natural), urban infrastructure (built), and population vulnerabilities (social). Similarly, Aquilino et al. (2021) enhance urban sustainability indicators by combining tri-environmental data to improve population mapping accuracy, allowing more precise evaluations of who might be affected by environmental hazards (social). Comparable efforts in other domains reinforce its versatility: Koks et al. (2019) demonstrate a global multi-hazard framework that integrates hazard interactions (natural), infrastructure exposure (built), and societal dependence (social), while Chang et al. (2021) employ the Social–Ecological–Technological Systems (SETS) framework to assess urban flood vulnerability across six U.S. cities by linking flooding dynamics (natural), physical infrastructure (built), and community vulnerability (social). Collectively, these examples demonstrate the value of integrated spatial analyses for urban planning and risk assessment. However, most existing applications concentrate on floods, heatwaves, or other climate-related hazards, and they typically operate at static or coarse temporal scales (Li et al., 2024).

In wildfire research, tri-environmental applications remain limited but are gaining traction. Nunes et al. (2024) develop a settlement-scale evacuation-priority model in Portugal that explicitly integrates wildfire hazard (natural), evacuation networks and shelter infrastructure (built), and demographic vulnerability (social). Johnson et al. (2023) combine mechanistic fire simulations (natural) with agent-based landowner decision-making (social) in Oregon, illustrating how land development (built) and community behavior (social) interact with natural fire dynamics. Moritz et al. (2022) advance a regional wildfire mitigation program that goes beyond fuel reduction (natural) to include retrofitting structures (built), reinforcing evacuation capacity (built), and building community preparedness (social). Reviews further highlight gaps: while hazard–built integration is progressing (natural + built), social vulnerability metrics (social) remain underrepresented, and key drivers such as ember transport (natural), intra-community spread (built), and adaptive capacities like insurance or warning compliance (social) are seldom included (Dunn et al., 2020).

This study builds on prior work by advancing the tri-environmental framework through its implementation at a daily temporal resolution for wildfire analysis. By capturing how the natural, built, and social interact and evolve during the course of a single fire event, our approach moves beyond compartmentalized or retrospective assessments. Such an integrated perspective enhances the precision of risk assessments and informs the development of targeted mitigation strategies to support resilience in wildfire-prone urban environments.

3. Study area and data

3.1. Study area

Eaton and Palisades (Fig. 1) are two wildfire-prone regions located in Los Angeles County, California. Both represent distinct examples of WUI environments, where human development intersects with fire-adapted natural landscapes. These areas are characterized by complex terrain, diverse vegetation assemblages, and a high frequency of wildfire exposure (Radeloff et al., 2018).

Eaton (Fig. 1, bottom right), located along the foothills of the San Gabriel Mountains, is characterized by steep canyons, dense evergreen forests, and extensive chaparral. Medium- to low-density residential development interspersed with wildland vegetation forms a classic intermix WUI. Wildfire vulnerability is heightened by Santa Ana winds that funnel through canyons, accelerating downslope fire spread, and by limited evacuation infrastructure (Abatzoglou et al., 2023). The spatial proximity of flammable vegetation and housing complicates suppression and emergency response. Palisades (Fig. 1, top right) occupies the coastal slopes of the Santa Monica Mountains, where a Mediterranean climate with prolonged dry summers and offshore winds, creates conditions favorable for ignition and rapid fire spread. The area comprises a mosaic of residential zones, wildland parks (e.g., Topanga State Park),

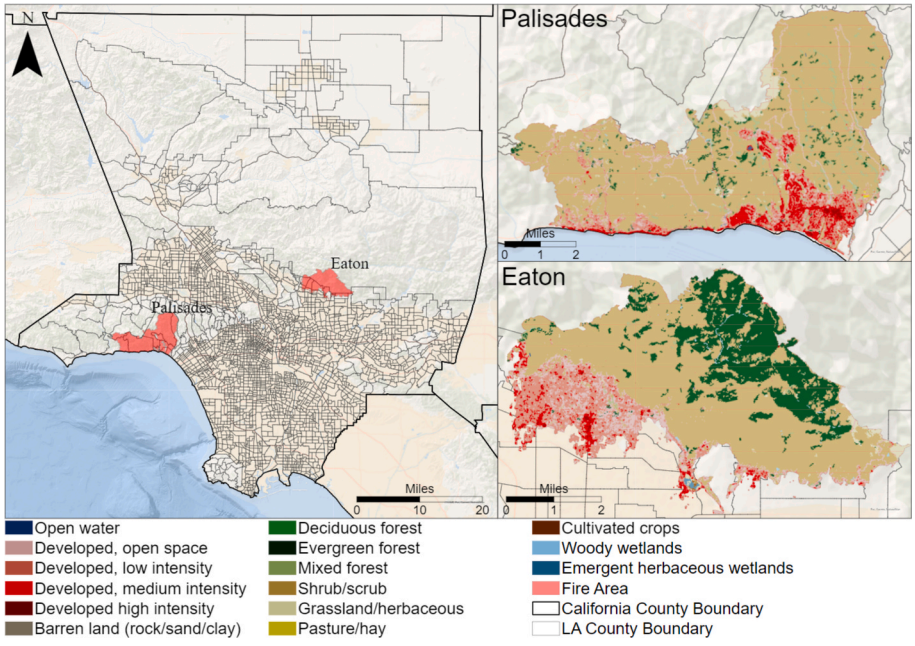


Fig. 1. Wildfire-Affected Areas in Palisades and Eaton, Los Angeles. Note: The left panel shows the locations of Palisades and Eaton in Los Angeles County. The right panels display enlarged land cover and burned areas in each fire zone.

and undeveloped shrublands. Steep terrain and dense housing adjacent to vegetated hillsides form a continuous fuel matrix across the WUI (Syphard et al., 2012). Both regions face increasing wildfire frequency and intensity due to climate-driven aridification, fuel accumulation, and the proliferation of invasive grasses, highly flammable surface fuels that exacerbate fire behavior amid continued urban expansion (Balch et al., 2022; Modaresi Rad et al., 2023).

3.2. Data collection and manipulation

To assess the natural, built, and social environments of the 2025 Los Angeles wildfire, we construct a comprehensive spatial dataset library that integrates fire boundaries, thermal activity, land cover, demographic, and POI information. These datasets are selected based on spatial resolution, update frequency, and thematic relevance to wildfire impact analysis (Table 1).

Fire Boundary and Thermal Activity. The wildfire perimeter dataset from the California Department of Forestry and Fire Protection (CAL FIRE) delineates the official burn extent from July 1 to January 13, 2025 (CAL FIRE, 2025). To capture temporal dynamics, this study uses daily thermal anomaly data from the VIIRS sensor aboard NOAA-21. VIIRS provides 375-meter resolution imagery based on mid-infrared signatures and supports near-real-time detection of active fire fronts (Schroeder et al., 2014).

Natural Environment. To represent the natural environment, we collect and apply the 2021 National Land Cover Database (NLCD) (Dewitz, 2023), produced by the U.S. Geological Survey. The NLCD provides a 30 m resolution raster classification of land cover types nationwide, distinguishing forests, shrublands, grasslands, wetlands, and developed areas.

Built Environment. To represent the built environment dimension, the study uses OpenStreetMap (OpenStreetMap, 2024) building footprints and road networks, together with Foursquare Open Source Places (2024) points of interest (POIs). OSM captures the physical infrastructure of the built environment, including roads and buildings that are critical to wildfire exposure and accessibility. Foursquare POIs extend this representation to the functional dimension, identifying hospitals, schools, retail centers, and public services as community activity hubs.

Social Environment. To represent the social environment dimension, we use U.S. Census Bureau (2020) data at the census tract level. Census tract attributes such as age, income, and race capture social environment characteristics.

Potential Economic Exposure Estimation. To translate exposure into potential economic exposure, we use standardized baseline unit values for land use, roads, and buildings (Table 2), derived from established industry and governmental sources. Recognizing that asset values exhibit substantial spatial heterogeneity and valuation uncertainty, particularly across urban environments, these baseline prices are complemented by asset-class–specific uncertainty ranges. Following established practice in disaster risk and infrastructure assessment, uncertainty ranges are applied uniformly within each major asset class to reflect plausible variability in market values and replacement costs (Koks et al., 2019). Specifically, a -40% to $+60\%$ range is adopted for land-use categories to capture documented intra-urban land price heterogeneity, following similar study done by Pavesi et al. (2024). A $\pm$ 30% range is applied to building replacement costs, consistent with FEMA HAZUS guidance and reported RSMeans cost variability (FEMA, 2020). For roads and transportation infrastructure, a $\pm$ 40% range is used to reflect scenario-based cost bands reported in federal and state transportation cost guides (FHWA, 2022; Caltrans, 2021). Together, these ranges enable the estimation of low, baseline, and high potential exposure scenarios, providing a transparent representation of valuation uncertainty without requiring parcel-level hedonic pricing or project-specific cost modeling.

4. Methodology

This study adopts a multi-step geospatial methodology to assess wildfire exposure across natural, built, and social environments in two WUI zones in Los Angeles County. The framework integrates multiple data sources—thermal satellite imagery, official fire boundaries, land cover classifications, infrastructure layers, and dasymetrically mapped population data—into a spatially explicit and temporally dynamic assessment. The workflow consists of three core components: (1) delineation of daily wildfire extents, (2) downscaling of population density using dasymetric mapping, and (3) integrated impact evaluation through a tri-environmental lens, as illustrated in Fig. 2.

4.1. Wildfire area identification

To accurately assess the spatial extent and temporal progression of the 2025 Los Angeles wildfire, we implemented a hybrid framework that integrates official fire perimeter data with satellite-derived thermal observations (Fig. 2, Step 1). The static fire boundary was obtained from CAL FIRE, delineating the overall extent of the burned area from July 1 to January 13, 2025. While this vector dataset provided a reliable outline of cumulative fire impact, it lacked the temporal granularity needed to track daily fire dynamics.

To capture day-by-day fire activity within this perimeter, we incorporated thermal anomaly data from the VIIRS aboard the NOAA-21 satellite. This dataset delivered 375-meter resolution nighttime imagery, enabling high-frequency detection of active fire fronts based on mid-infrared radiative signals. We applied KDE to the VIIRS point detections, generating a continuous surface that highlights zones of concentrated heat activity and frequent ignition. The resulting raster was thresholded to distinguish burn-affected areas and subsequently converted into vector polygon format, producing a sequence of daily fire boundaries. By intersecting these daily polygons with the CAL FIRE perimeter, we generated refined, temporally disaggregated fire maps that integrated both satellite-detected hotspots and officially reported

Table 1
Sources of Datasets Included in the Spatial Data Layers Library.

Attribute	Los Angeles fire boundary	VIIRS NOAA-21	OSM Data	Foursquare Open Source Places	Land Cover Data	Census Data
Category	Fire Information	Daily wildfire spread	Built Environment	Built Environment	Natural Environment	Social Environment
Source	CAL FIRE	NASA	OpenStreetMap	Foursquare	USGS/NLCD	US Census Bureau
Format	Vector	Raster	Vector	POI Data	Raster	Vector
Spatial Resolution	N/A	375 *m*	N/A	Global Level	30 *m*	County level
Temporal Resolution	N/A	Daily	N/A	Annual	Annual	Decennial
Time	07/01/2025–13/01/2025	07/01/2025–13/01/2025	2024	2024	2021	2020

Note: CAL FIRE: California Department of Forestry and Fire Protection; VIIRS: Visible Infrared Imaging Radiometer Suite; NOAA: National Oceanic and Atmospheric Administration; OSM: OpenStreetMap; NLCD: National Land Cover Database; USGS: United States Geological Survey.

Table 2
Asset Unit Values Applied in Tri-Environmental Wildfire Exposure Assessment.

Asset Class	Subcategory	Unit	Baseline Price (USD)	Uncertainty Range	Source
Land Use	Residential land	Acre	8,536,453	−40% to + 60%	RS Means – Residential(Gordian, 2023)
	Industrial land	Acre	5,606,608	−40% to + 60%	RS Means – Factory(Gordian, 2023)
	Grassland	Acre	479,160	−40% to + 60%	Sports Venue Calculator
	Commercial land	Acre	5,679,353	−40% to + 60%	RS Means – Dept. Store (Gordian, 2023)
	Retail land	Acre	5,874,502	−40% to + 60%	RS Means – Retail (Gordian, 2023)
	Park	Acre	981,250	−40% to + 60%	MRG Solutions
	Cemetery	Acre	80,000	−40% to + 60%	Foresight Companies
	Farmland	Acre	5,586	−40% to + 60%	AcreTrader
	Forest	Acre	130	−40% to + 60%	CT.gov DOAG
	Orchard	Acre	750	−40% to + 60%	CT.gov DOAG
	Heath	Acre	1,480	−40% to + 60%	Agriculture.com
	Scrub	Acre	40	−40% to + 60%	CT.gov DOAG
Roads	Pedestrian paths	Mile	0.75 M	±40%	FHWA / Caltrans (Federal Highway Administration, 2022)
	Residential & service roads	Mile	3.5 M	±40%	FHWA / FEMA (Federal Highway Administration, 2022)
	Primary roads	Mile	8.5 M	±40%	FHWA / FEMA (Federal Highway Administration, 2022)
	Tracks / Unclassified	Mile	2.0 M	±40%	FHWA / DOT (Federal Highway Administration, 2022)
Buildings	Residential (house, apartment, hut)	Sq. ft.	210	±30%	RS Means – Residential (Gordian, 2023)
	Commercial (retail, hotel, mall, bank, cafe, restaurant, supermarket, church, etc.)	Sq. ft.	215	±30%	RS Means – Commercial (Gordian, 2023)
	Industrial (factory/warehouse)	Sq. ft.	200	±30%	RS Means – Industrial (Gordian, 2023)

Note: Baseline unit values are derived from standard industry and governmental sources. Asset-class–specific uncertainty ranges (−40% to + 60% for land use, ±30% for buildings, and ± 40% for roads) are applied to reflect valuation uncertainty and support low–high exposure estimates.

perimeters.

Building on these refined daily fire boundaries, we assess wildfire loss in terms of potential economic exposure, defined as the maximum value of assets intersecting with the fire perimeter under the assumption of complete destruction. This exposure-based approach is widely used in wildfire risk research because it provides a consistent and operational proxy for potential damages, even when detailed severity or fire behavior data are unavailable. For example, Monge et al. (2023) integrate wildfire, climate, and land-use forecasts into probabilistic models to project economy-wide impacts, effectively treating exposure as a core component of potential loss, while Bergonse et al. (2022) demonstrate that exposure contributes substantially to local-scale wildfire risk assessments in Portugal. While realized damages are further conditioned by burn severity, flame intensity, and suppression effectiveness, those factors are event-specific and often lack consistent spatial and temporal coverage. Using potential exposure therefore enables robust comparisons across regions and time periods, and provides a baseline that can be extended with severity indices or fire behavior models in future studies.

Thus, these refined daily fire boundaries serve as the spatial foundation upon which the subsequent tri-environmental framework is applied to evaluate wildfire impacts on the natural, built, and social environments.

4.2. Downscaling population through dasymetric mapping

Conventional gridded population datasets such as LandScanand WorldPop (2021) offer valuable population estimates, but their spatial resolutions (1 km and 100–1000 m, respectively) are insufficient for detailed wildfire impact analysis at the neighborhood or WUI scale. To overcome this limitation, we employed a dasymetric mapping approach that downscaled population data to a 20 m × 20 m resolution, allowing more precise identification of exposed populations.

Dasymetric mapping is a spatial redistribution method that disaggregates census-level population data using land cover and ancillary variables to reflect the physical plausibility of habitation (Mennis, 2003; Leyk et al., 2019). In our study, we incorporated 2020 U.S. Census block-level data and the 2021 NLCD to derive population density surfaces that account for heterogeneity in land use across Eaton and Palisades.

The dasymetric mapping process (Fig. 2, Step 2) involved a multi-step workflow to redistribute population data based on land use suitability. First, we assigned Relative-weight Values (RA) to each land cover class according to its potential to host human populations. Developed areas—particularly those with high residential intensity—received higher weights, while natural features such as forests, wetlands, and open water were assigned minimal or zero values (Table 3). These RA values were used to generate the Population Allocation Factor Raster, as expressed in Equation (1):

$$RA = f(\text{LandUseType}) \tag{1}$$

Next, we used the Tabulate Area tool to calculate the total number of pixels associated with each land cover class in each census block, as shown in Equation (2):

$$\text{TotalPixel} = \sum(\text{PixelCount}) \tag{2}$$

These values allowed us to compute the proportion of each land cover type within a block and estimated the expected population for each class accordingly. Finally, we normalized and adjusted population values using the Adjusted Population Formula in Equation (3):

$$\text{AdjustedPopulation} = \frac{RA \times Pop \times 400}{\text{TotalPixel} \times \text{ExpectedPopulation}} \tag{3}$$

Here, Pop was the total census population for a block, RA was the relative weight for the given land cover class, and 400 represents the area in square meters of each raster cell (20 m × 20 m). This adjustment step ensured that the population was distributed in accordance with the physical land suitability and spatial heterogeneity across the WUI landscape.

4.3. Tri-environmental framework for fire impact evaluation

To comprehensively evaluate the multidimensional effects of the

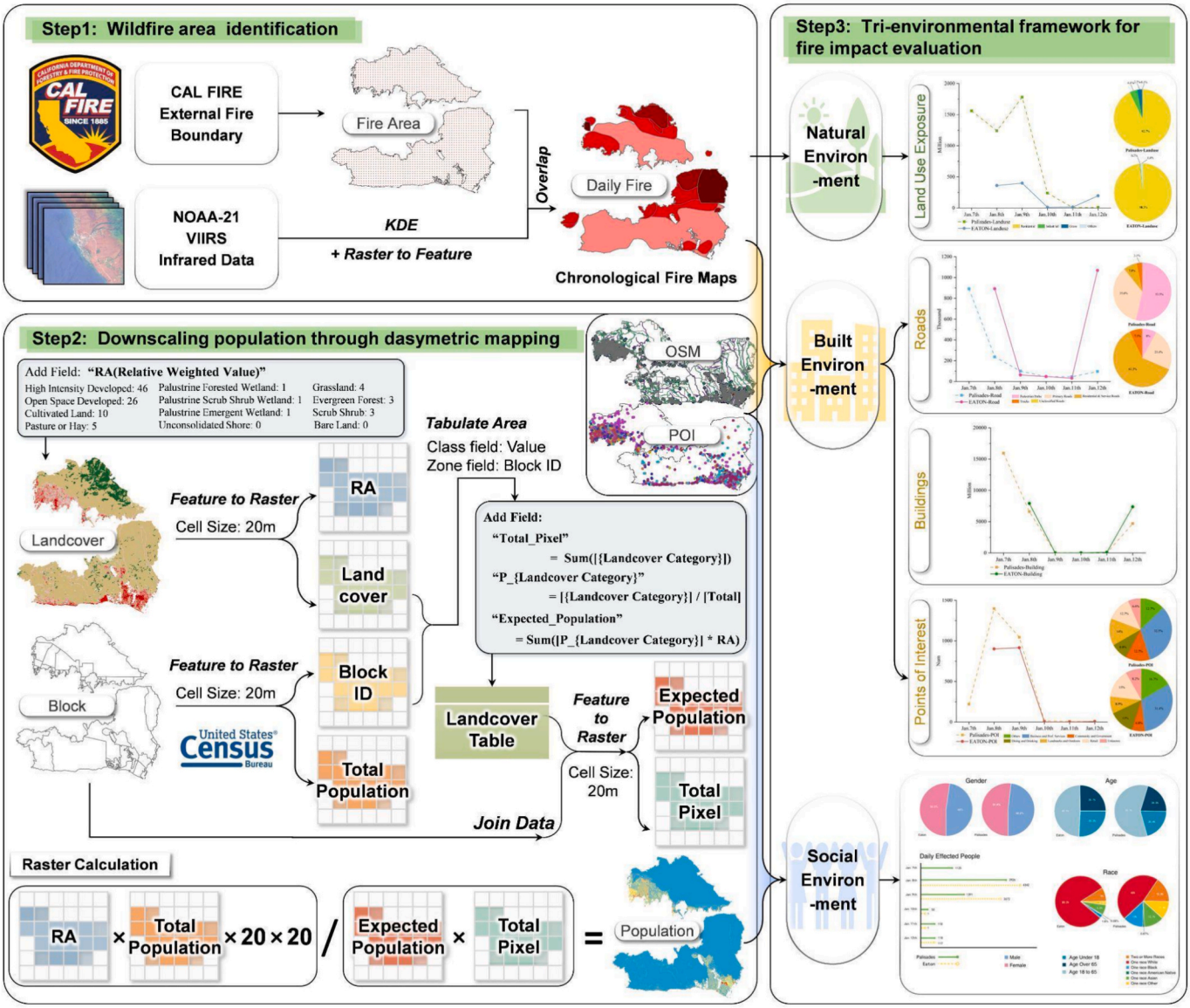


Fig. 2. Workflow of the Tri-Environmental Wildfire Impact Assessment Framework.

Table 3
Relative Weight Values Assigned to Land Cover Types for Dasymetric Mapping.

Land Cover Type	Relative Weighted Value (RA)
Open water	0
Developed, open space	26
Developed, low intensity	10
Developed, medium intensity	15
Developed, high intensity	46
Barren land	0
Deciduous forest	3
Evergreen forest	3
Mixed forest	4
Shrub/scrub	3
Grassland/herbaceous	4
Pasture/hay	5
Cultivated crops	10
Woody wetlands	1
Emergent herbaceous wetlands	1

Note: Relative-weighted Values (RA) reflected the suitability of each land cover type for population distribution, with higher weights for developed areas and lower or zero weights for natural or uninhabited land.

2025 Los Angeles wildfire, we implemented a tri-environmental framework that assessed exposure across three interconnected domains: the natural environment, the built environment, and the social environment (Fig. 2, Step 3). This integrative framework enabled a holistic understanding of wildfire exposure patterns by combining tri-environmental datasets within the defined fire-affected zones.

Natural and Built Environment Assessment. We began by overlaying the refined wildfire polygons (derived in Section 4.1) with vector-based datasets from OpenStreetMap (OpenStreetMap, 2024), which provided detailed geospatial representations of building footprints, road networks, and landscape features such as parks and vegetated areas. This spatial intersection allowed us to identify natural and built environment types that fell within the fire perimeter, offering direct evidence of potential economic exposure. To assess disruptions to community-serving built environment, we also incorporated Foursquare Open Source Places, which catalogued a wide array of POIs across Los Angeles, including hospitals, schools, retail businesses, and restaurants. Intersecting POI locations with the wildfire boundary enabled us to quantify functional losses and evaluate the exposure of critical urban amenities to wildfire threats.

Social Environment Assessment. To analyze the fire's social

exposure, we intersected the daily wildfire footprint with the high-resolution population density surface generated through dasymetric mapping (Section 4.2). This overlay allowed for precise estimation of the number of residents exposed to fire hazards on a per-pixel basis, rather than relying on coarse administrative units. We further enriched this spatial analysis by integrating 2020 U.S. Census demographic data, which enabled disaggregation of affected populations by age group, gender, and income level at the census tract scale. Through this process, we identified vulnerable subpopulations—such as children, elderly individuals, and low-income households—residing within or adjacent to the fire-affected zones. By linking demographic profiles with spatial exposure data, this framework highlighted patterns of social vulnerability that were critical for informing targeted evacuation, relief distribution, and long-term recovery planning.

Potential Economic Exposure Estimation. To quantify wildfire-related economic exposure within the natural and built environments, we defined "loss" as potential economic exposure. For each day d, exposure was estimated as the sum of land use, road, and building values intersecting the fire footprint, as shown in Equation (4):

$$Loss_d = Loss_d^{LU} + Loss_d^{RD} + Loss_d^{BD} \tag{4}$$

where $Loss_d^{LU}$ denoted land-use exposure, $Loss_d^{RD}$ denoted road exposure, and $Loss_d^{BD}$ denoted building exposure. Each component was derived by multiplying the exposed quantity of an asset by its corresponding unit value.

For land use, exposure was calculated as shown in Equation (5):

$$Loss_d^{LU} = \sum\nolimits_{c \in C_{LU}} A_{c,d} \times V_c \tag{5}$$

where $A_{c,d}$ was the burned area (acres) of land-use type c on day d, and V_c was its per-acre unit value (USD/acre).

For roads, exposure was calculated as shown in Equation (6):

$$Loss_d^{RD} = \sum\nolimits_{t \in T_{RD}} L_{t,d} \times C_t \tag{6}$$

where $L_{t,d}$ was the exposed road length (miles) of type t, and C_t was its per-mile replacement cost (USD/mile).

For buildings, exposure was calculated as shown in Equation (7):

$$Loss_d^{BD} = \sum\nolimits_{s \in S_{BD}} S_{s,d} \times U_s \tag{7}$$

where $S_{s,d}$ was the exposed building footprint (sq.ft.) of type s, and U_s was its per-square-foot replacement cost (USD/sq.ft.).

Unit values were drawn from standardized sources (e.g., RS Means, FEMA HAZUS, FHWA) and sector-specific valuation reports (Table 2), ensuring comparability between asset classes. Within the tri-environmental framework, this calculation represented the maximum potential exposure-based loss, serving as an upper-bound estimate of economic risk since it did not incorporate factors such as fire suppression, construction resilience, or post-event repair trajectories. By embedding these estimates in the tri-environmental framework, the study directly linked spatial exposure with monetary values, thereby bridging fine-scale hazard mapping and economic consequence assessment and setting the foundation for cross-comparisons with broader economic loss studies.

Cross-Comparative Economic Loss Estimates with Other Assessments. To evaluate the validity and applicability of our potential economic exposure framework, we incorporated a cross-comparative design with two regional-scale assessments of the 2025 Los Angeles wildfire: the UCLA Anderson Forecast and the Los Angeles County Economic Development Corporation (LAEDC) report. This comparative structure enabled us to assess the internal consistency of our estimates and explore how methodological differences—such as scale, input data, and attribution logic—could influence the interpretation of wildfire-related economic consequences. Although the focus of this study remained on localized, asset-level potential exposure, aligning our estimates with broader benchmarks provided an important lens for understanding both the strengths and the boundaries of spatially disaggregated disaster impact models.

5. Results

5.1. Wildfire progression and spatial dynamics

The spatiotemporal dynamics of wildfire progression in both Eaton and Palisades are illustrated in Fig. 3, which displays daily burned areas and aggregated ignition points from January 7 to January 12, 2025. This visual record, based on combined CAL FIRE boundaries and VIIRS thermal detections, captures the evolving nature of fire behavior across two distinct topographic contexts within Los Angeles County.

In Eaton, wildfire activity initiates on January 7 with localized ignitions on north-facing slopes of the San Gabriel Mountains, northeast of Mount Wilson. These early burns occur in steep, densely vegetated terrain conducive to fire spread. By January 8, the fire advances southeast along the Altadena–Chaney Trail corridor toward the urban fringe. On January 9, it undergoes substantial expansion, forming a continuous burn front from eastern Altadena toward Sierra Madre. A new ignition on January 10 near Millard Canyon, possibly caused by ember spotting, adds spatial complexity. Activity becomes fragmented by January 11, with scattered clusters moving southwest and upslope. On January 12, only low-intensity residual burning persists near Las Flores Canyon. The cumulative burn scar spans rugged forested slopes and WUI edges, highlighting suppression challenges in topographically complex terrain.

In contrast, wildfire activity in the Palisades region is first observed on January 7 within Topanga State Park, near Entrada Road and Trippet Ranch, areas known for steep ridgelines and flammable chaparral. On January 8, it spreads bidirectionally, east toward Will Rogers State Park and west into Santa Ynez Canyon, driven by local wind and terrain dynamics. Peak activity occurs on January 9, forming a continuous ignition belt across Tuna Canyon and Los Liones Trail, threatening ridgeline residences. Between January 10–11, the fire progresses along Temescal Ridge and Rustic Canyon, with irregular daily spread indicative of intense flanking and potential crown fire behavior. By January 12, only minor hotspots remain in Rivas and Hondo Canyons. The final burn footprint reveals dense ignition patterns across natural and developed areas, underscoring high WUI exposure and ecological vulnerability.

5.2. Impact analysis for the natural & built environment

Wildfires in Eaton and Palisades have markedly different exposures across both natural and built environments. Through the integration of temporal fire boundary data and spatial overlays with land use, infrastructure, and POI datasets, we evaluate daily patterns and cumulative consequences on critical landscape and human systems. Detailed numerical results of exposure and potential economic exposure estimates can be found in Appendix A.

5.2.1. Natural Environment: Land use exposure

Aggregated over the study period (January 7–12), wildfire exposure to natural and residential land uses differs markedly between the two study areas (Fig. 4), with composition shown as stacked area charts in Fig. A1. In Palisades, cumulative land-use–related potential economic exposure reaches a baseline total of approximately USD 4.8 billion, with an uncertainty range of USD 2.9–7.7 billion, reflecting a − 40% to + 60% adjustment applied to land-use valuation. In Eaton, the cumulative exposure is substantially lower, with a baseline estimate of approximately USD 1.1 billion and a corresponding uncertainty range of USD 0.6–1.6 billion. These ranges explicitly account for spatial heterogeneity in land values and are intended to represent upper- and lower-bound

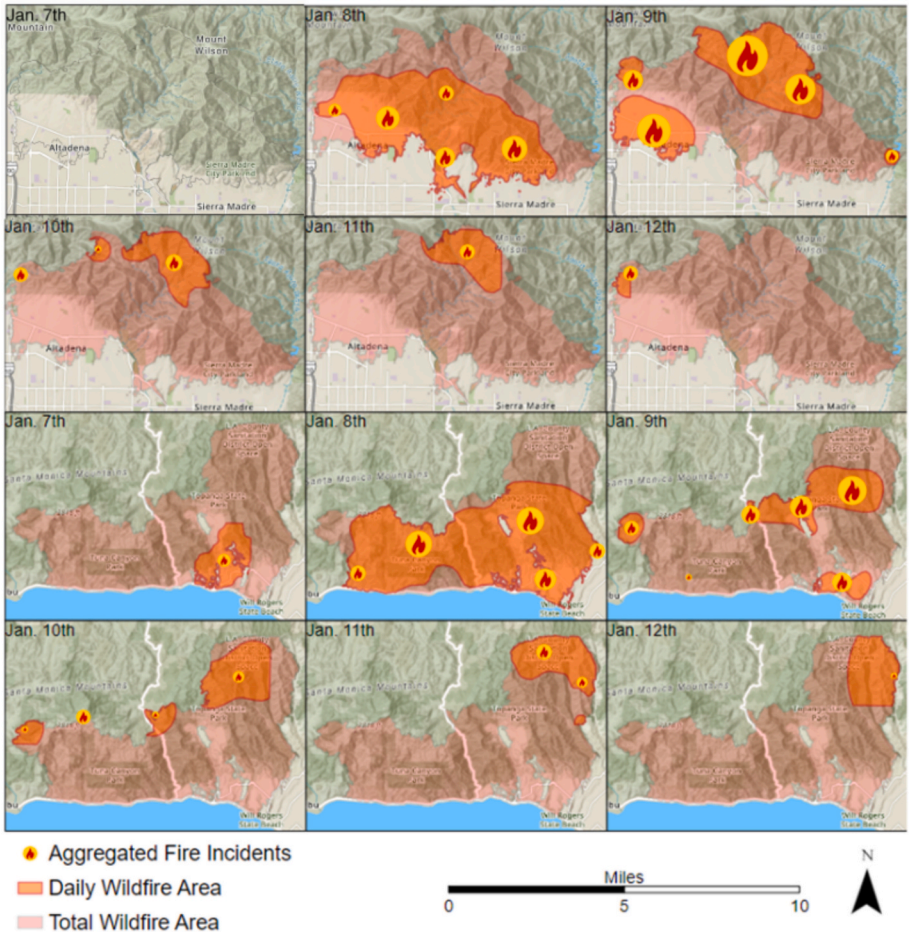


Fig. 3. Daily Progression of Wildfire Spread in Eaton and Palisades (January 7–12, 2025). Note: The figure shows daily fire spread in Eaton (top six maps) and Palisades (bottom six maps) using VIIRS thermal detections and CAL FIRE perimeters. Orange polygons indicate new burns; pink shading shows cumulative extent; flame icons mark daily hotspots (Eaton: January 8–12; Palisades: January 7–12, For Eaton, the Jan 7 panel is shown as a zero baseline (no fire activity/perimeter); the fire ignited on Jan 8.). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

estimates of short-term potential exposure, rather than realized post-fire economic losses.

Temporally, land-use exposure in both regions is strongly concentrated during the early phase of fire progression, with daily exposure peaking on January 9 (Fig. 4). In Palisades, the peak daily exposure reaches a baseline of approximately USD 1.8 billion (uncertainty range: USD 1.1–2.8 billion), while Eaton exhibits a lower peak of about USD 0.4 billion (uncertainty range: USD 0.24–0.64 billion). Exposure declines sharply after January 10 in both regions, indicating that cumulative land-use exposure is dominated by a narrow temporal window of active fire spread. Despite similar temporal dynamics, the composition of exposed land uses differs substantially. Exposure in Eaton is overwhelmingly residential (98.32%), with minimal contributions from grass (1.42%) and industrial or other land uses (0.26%). In contrast, Palisades shows a more heterogeneous profile, with residential land accounting for 92.73%, alongside higher exposure shares from grass (4.43%) and industrial or open land (2.69%), reflecting wildfire impacts across a broader set of ecological and residential–vegetation interface zones.

5.2.2. *Built Environment: Roads*

Over the full fire progression period (January 7–12), wildfire exposure to road infrastructure results in notable cumulative impacts in both study areas (Fig. 5), with composition shown as stacked area charts in Fig. A2. Under the baseline scenario, total road-related potential economic exposure in Palisades is estimated at approximately USD 1.41 million, obtained by aggregating daily exposures across the event window. Incorporating the ± 40% uncertainty range for road valuation yields a total exposure interval of USD 0.85–1.98 million, reflecting variability in road construction standards and replacement costs.

Temporally, road exposure in both regions is concentrated during the early phase of fire spread, with elevated values observed between January 7 and January 9, followed by a rapid decline after January 10 as fire activity subsides. While daily exposure levels in Eaton are generally lower than those in Palisades during the initial days, cumulative effects remain substantial due to sustained exposure across multiple dates. In terms of composition, the exposed road network differs markedly between the two regions. In Eaton, exposure is dominated by residential and service roads, accounting for approximately 61.5% of total road exposure, followed by primary roads (23.4%), with smaller contributions from pedestrian paths and tracks. In contrast, Palisades shows a more diversified structure, where pedestrian paths (53.5%) and primary roads (35.6%) together constitute the majority of exposed road infrastructure, indicating greater exposure along mixed-use and pedestrian-oriented networks.

5.2.3. *Built Environment: Buildings*

As shown in Fig. 6, building-related potential economic exposure is substantial in both regions, with timing and intensity reflecting differences in fire entry into built environments. Over the full study period, baseline estimates show higher cumulative building exposure in Palisades than in Eaton. To account for uncertainty in building replacement

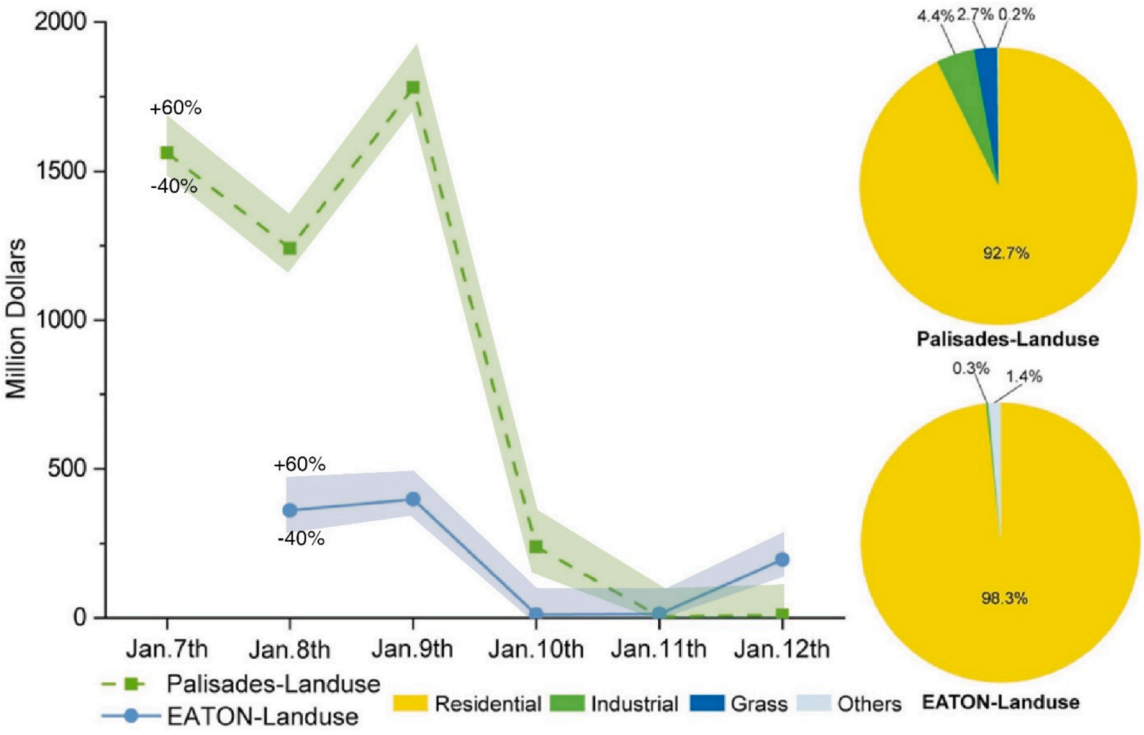


Fig. 4. Temporal Changes and Composition of Land Use Affected by Wildfire. Note: Solid lines represent baseline estimates, while shaded envelopes indicate uncertainty ranges derived from land-use valuation variability (−40% to + 60%). Pie charts summarize cumulative exposure composition by land-use category.

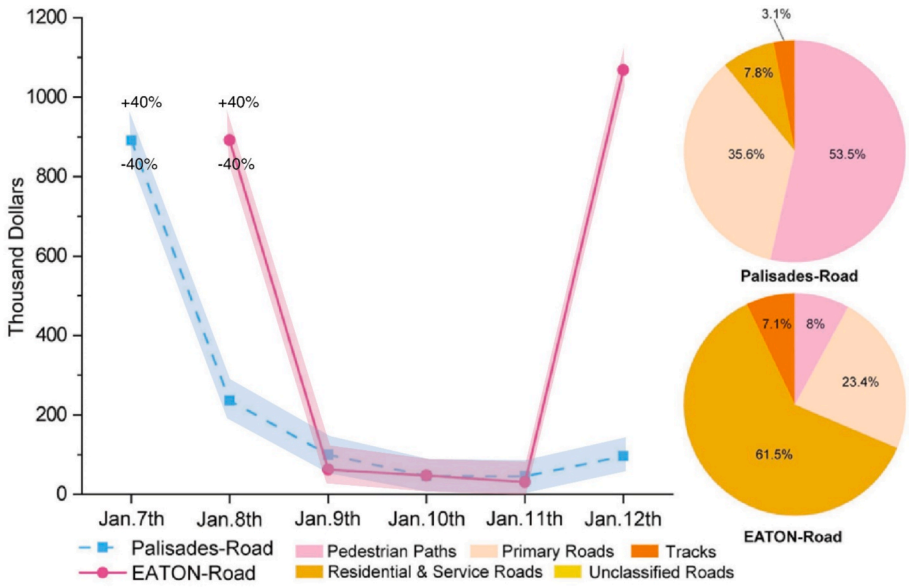


Fig. 5. Temporal Changes and Composition of Road Exposed by Wildfire. Note: The line chart shows daily road exposure estimates, while shaded envelopes indicate uncertainty ranges derived from variability in road asset valuation (−40% to + 40%); pie charts depict exposed road types, with pedestrian paths dominant in Palisades and residential/service roads in Eaton.

costs, all estimates incorporate a ± 30% uncertainty range, yielding cumulative exposure ranges of approximately USD 11.2–20.7 billion in Palisades and USD 5.5–10.3 billion in Eaton.

Temporally, building exposure in both regions is concentrated during the early phase of fire progression. In Palisades, daily exposure peaks on January 7, with a baseline value of approximately USD 16.0 billion, followed by a rapid decline after January 8. Eaton exhibits a similar temporal pattern at a lower magnitude, reaching its maximum daily exposure on January 8 with a baseline estimate of approximately USD 7.9 billion. Daily exposure decreases substantially in both regions after January 9, indicating that most building-related exposure occurs within a short time window during active fire spread.

5.2.4. *Built Environment: Points of interest (POIs)*

POIs exposure patterns closely mirror wildfire advancement. As depicted in Fig. 7, Eaton records its highest number of affected POIs on January 8 (1,397), while Palisades peaks on January 9 (915). After January 10, both regions show a sharp decline, reflecting either

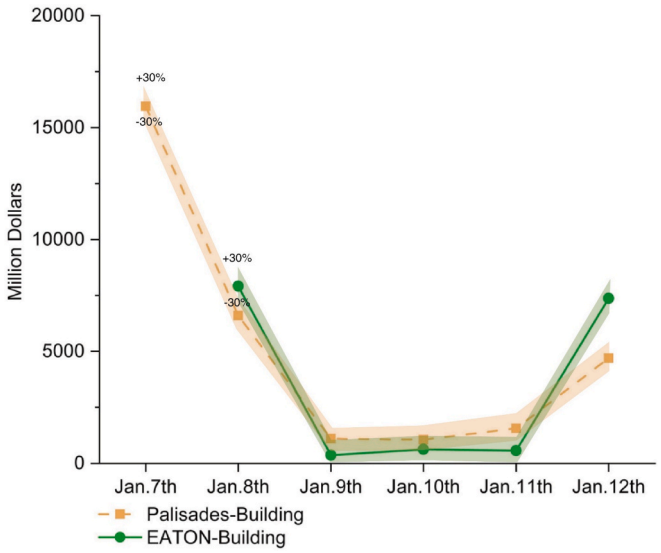


Fig. 6. Daily Economic Losses from Fire-Damaged Buildings. Note: The line chart shows daily estimated potential economic exposure from buildings in Palisades and Eaton, while shaded envelopes indicate uncertainty ranges derived from variability in road asset valuation (−30% to + 30%).

successful containment or the fire's retreat from core urban areas.

POI type distributions further highlight differences in community exposure. In Eaton, the largest affected categories are Business and Professional Services (34.74%), followed by Retail (14.99%), Dining and Drinking (13.55%), and Community and Government services (13.64%). In Palisades, POIs are more evenly distributed, with Business Services (24.08%), Community and Government (12.76%), and Retail (11.46%) making up the majority. This indicates that Eaton's exposure is more economically focused, while Palisades involves a broader range of public-facing and mixed-use functions.

5.3. Impact analysis for the social environment

To assess the social exposures of the 2025 Los Angeles wildfire, we intersect high-resolution population density layers, generated through dasymetric mapping with daily wildfire perimeters. This approach enables fine-scale analysis of both the temporal progression and demographic composition of exposed populations in Eaton and Palisades—two WUI communities with distinct vulnerability profiles.

As illustrated in Fig. 8, the dasymetric population density surfaces highlight sharp variations in population distribution. Warmer colors (e.g., red) denote densely populated zones concentrated near residential developments, whereas cooler tones (blue and green) indicate lower-density or uninhabited regions dominated by natural land cover.

Fig. 9 illustrates daily exposure trends and demographic compositions for Eaton and Palisades, revealing clear spatial and temporal differences. In Palisades, population clusters align with coastal residential areas and developed park-adjacent neighborhoods. In Eaton, the highest population densities are observed along the urban-wildland fringe, particularly near transportation corridors and suburban centers. These patterns validate the effectiveness of dasymetric redistribution in capturing real-world settlement structures and identifying socially vulnerable zones during fire events.

Daily population exposure patterns vary markedly between the two regions. Eaton experiences its peak on January 8, with 4,342 residents exposed—the highest single-day impact across the study period—followed by a sharp decline after January 10, indicating a brief but intense threat. In contrast, Palisades peaks on January 8 and 9 (3,926 and 1,391 residents, respectively), with exposure persisting at low levels through January 12, reflecting a more prolonged and diffuse impact.

Demographic profiles highlight differentiated vulnerabilities. Both areas exhibit near-balanced gender distributions, with a slight female majority (52.3% in Eaton, 51.8% in Palisades), underscoring the need for gender-responsive evacuation and recovery planning. Working-age adults (18–65) constitute the largest group in both regions, but Eaton has a notably higher proportion of seniors (25.7% vs. 20.5%), implying increased mobility and health-related risks. The combined presence of children and elderly in Eaton calls for accessible shelters and medical services. Racial composition further shapes risk profiles. Eaton's affected population is predominantly White (80.3%), while Palisades is more diverse—46% White, 15.8% Asian, 13% multiracial, and smaller shares of other groups—suggesting a greater need for multilingual and culturally adaptive communication strategies.

These social metrics, daily exposure trends, gender balance, age structure, and racial diversity collectively shape each community's resilience and risk profile. The data reinforces the necessity for equity-centered wildfire response planning. For Eaton, concentrated and high-intensity exposure necessitates rapid deployment of resources to senior populations and dense residential zones. In Palisades, extended exposure across a demographically varied population calls for sustained and inclusive community outreach efforts.

5.4. Comparison of economic loss estimates with other assessments

To contextualize and validate the findings of our tri-environmental

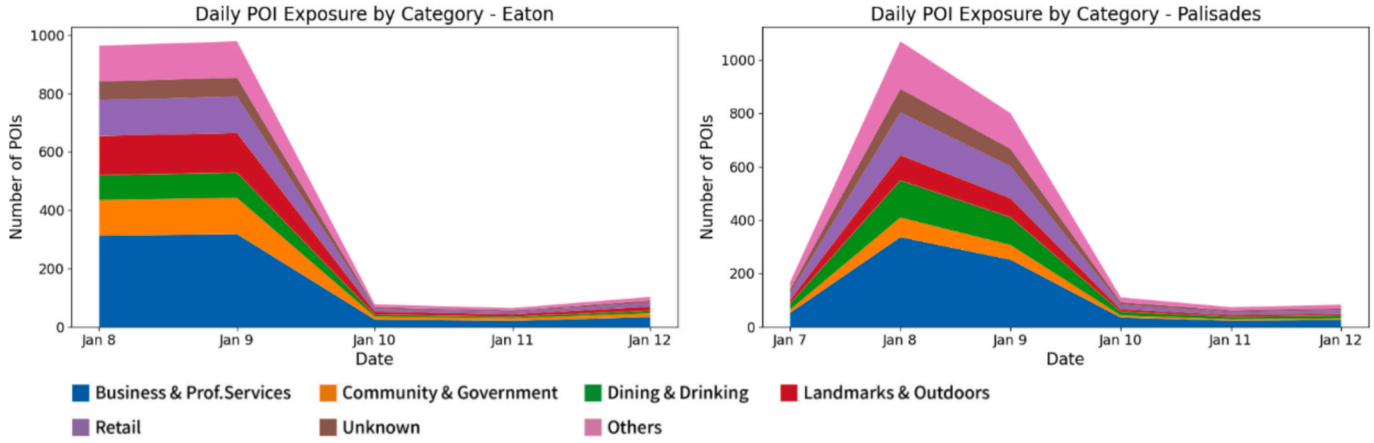


Fig. 7. Temporal Change and Type Composition of Fire-Affected POIs. Note: The stacked area charts show daily counts of fire-exposed Points of Interest (POIs) by category in Palisades and Eaton. "Unknown" indicates POIs without reliable category information in the original dataset, while "Others" includes rare or ambiguous categories that could not be clearly classified.

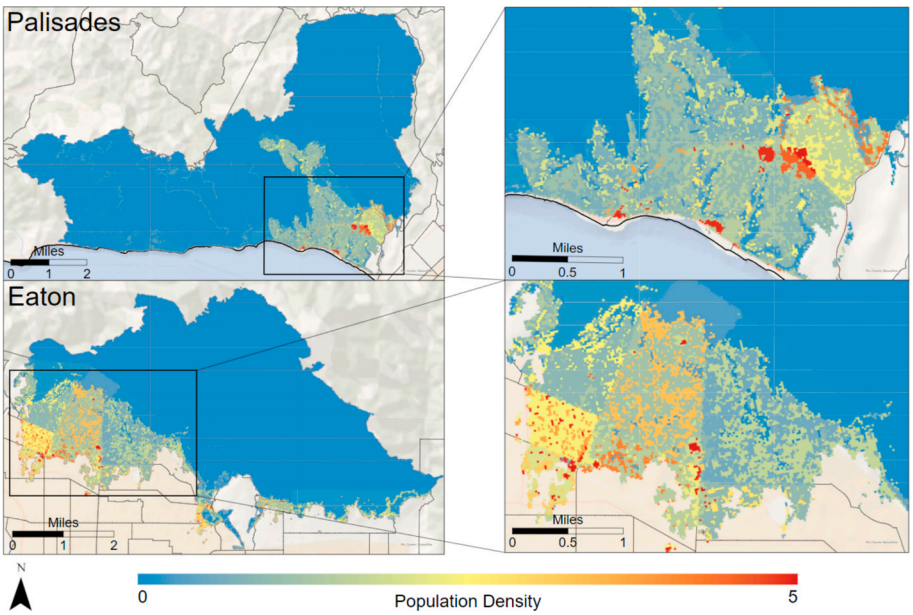


Fig. 8. Population density of Palisades and Eaton, Los Angeles by dasymetric mapping. Note: Warmer colors indicate higher population concentrations. Insets highlight detailed variations in population distribution, where high-density zones (in red) cluster near developed areas, and lower-density regions (in blue and green) align with open space and natural land cover. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

framework, we compare our localized potential economic exposure estimates with two prominent regional-scale assessments: the UCLA Anderson Forecast and the Los Angeles County Economic Development Corporation (LAEDC) report (UCLA Anderson, 2025; LAEDC, 2025). While all three studies assess wildfire-related economic impacts, they differ fundamentally in objectives, spatial and temporal resolution, impact definitions, asset granularity, and treatment of recovery processes. Our framework estimates daily, asset-level potential exposure within two WUI districts—Palisades and Eaton—whereas UCLA Anderson and LAEDC employ broader, top-down approaches at the county or regional scale. Key methodological differences across the three approaches are summarized in Table 4.

Our study is designed to quantify direct, short-term potential exposure during fire progression, using explicit daily fire dynamics derived from VIIRS and linking asset values to spatiotemporally resolved exposure. As a result, our estimates should be interpreted as upper-bound short-term exposure, rather than realized economic losses. In contrast, UCLA Anderson (2025) aims to estimate total economic and capital losses, operating at the county scale and over event-to-multi-year horizons. Its framework aggregates property and capital assets and relies on insurance-based loss estimates combined with macroeconomic scaling and GDP models, implicitly capturing long-term recovery effects. LAEDC, (2025) further extends this perspective by modeling direct, indirect, and induced economic impacts using input–output frameworks and explicit recovery scenarios (e.g., HAZUS), with a temporal focus spanning multiple years (2025–2029).

To further clarify the substantive differences across approaches, we next describe the key quantitative estimates reported by each assessment in detail (Tables 5–7). Specifically, as shown in Table 5, LAEDC, (2025) reports key quantitative estimates based on a post-fire, event-level framework emphasizing regional economic disruption and recovery-related outcomes. LAEDC estimates total property damage within burn perimeters at approximately USD 28.0–53.8 billion, affecting 20,218 parcels and 1,863 businesses across Palisades and Eaton. Beyond direct property impacts, the assessment highlights broader economic consequences, including 9,610 jobs located within fire zones, lost economic output of USD 4.6–8.9 billion over 2025–2029, 24,990–49,110 job-years lost, labor income reductions of USD 1.9–3.7 billion, and tax revenue losses of USD 0.73–1.41 billion. Overall, LAEDC focuses on medium-term, system-wide economic effects, capturing direct, indirect, and induced impacts relevant to regional recovery planning rather than localized, short-term exposure during fire progression.

Table 6 summarizes the key quantitative estimates reported by UCLA Anderson (2025), which focuses on county-wide property and capital losses resulting from the wildfire event, with an emphasis on post-fire realized economic impacts. UCLA Anderson estimates total property damage on the order of USD 28.0–53.8 billion, with insured losses of approximately USD 27.07 billion and scaled total direct property losses of about USD 59.55 billion, derived from insurance claims and loss-scaling assumptions. In addition to direct property impacts, the assessment reports substantial broader economic consequences, including lost economic output of USD 4.6–8.9 billion over 2025–2029, 24,990–49,110 job-years lost, labor income reductions of USD 1.9–3.7 billion, and tax revenue losses of USD 0.73–1.41 billion. Overall, UCLA Anderson's assessment emphasizes longer-term, realized losses at the county scale, integrating insurance-based estimates with macroeconomic modeling to characterize sustained economic impacts beyond the immediate fire period.

Table 7 reports the key quantitative estimates derived from our study, which focuses on daily, localized potential exposure during active fire progression (January 7–12) within the Palisades and Eaton WUI neighborhoods. Our estimates quantify exposure across natural, built, and social environments, including property assets, infrastructure, points of interest, and population groups directly intersecting with fire perimeters. For the built environment, we estimate building exposure of USD 29.91–55.55 billion and road exposure on the order of USD 0.002–0.005 billion, reflecting upper-bound short-term exposure rather than realized damage. Social impacts are characterized through temporally resolved population exposure, with a total of 15,861 exposed residents across the two districts during the fire period, along with disaggregated indicators by gender, age vulnerability, and race. Unlike county-level assessments, our study does not model recovery, indirect effects, or long-term economic losses; instead, it provides a high-resolution snapshot of where and to what extent assets and populations were directly exposed during fire progression, offering

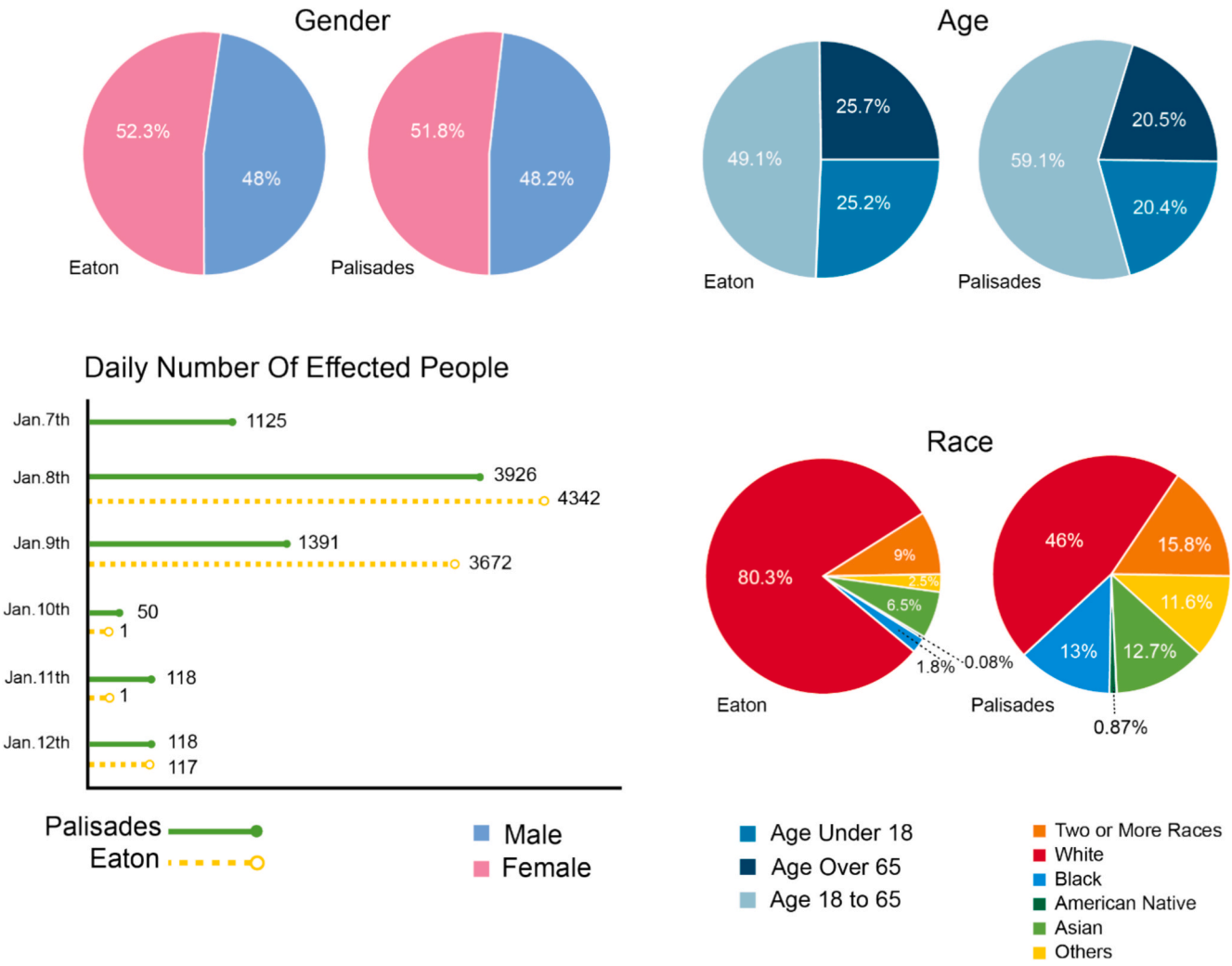


Fig. 9. Demographic Characteristics and Daily Population Exposure to Wildfire. Note: Pie charts depict gender, age, and racial composition in Eaton and Palisades. The bar chart shows the number of residents affected daily by wildfire from January 7–12, 2025.

Table 4
Cross-comparison of wildfire impact estimation approaches.

Dimension	Our Study	UCLA Anderson (2025)	LAEDC (2025)
Primary objective	Quantify daily, localized potential exposure	Estimate total economic & capital losses	Estimate regional economic impacts & recovery
Spatial scale	WUI neighborhood / asset-level (Palisades, Eaton)	Los Angeles County	Los Angeles County & SoCal region
Temporal resolution	Daily (Jan 7–12)	Event- to multi-year	Multi-year (2025–2029)
Impact type	Direct potential exposure only	Direct + indirect + long-term	Direct + indirect + induced
Asset granularity	Buildings, roads, POIs	Aggregated property & capital	Parcels & businesses
Fire dynamics	Explicit daily fire progression (VIIRS)	Not modeled	Not modeled
Economic mechanism	Asset value × daily exposure	Insurance losses × scaling + GDP models	Input–output (IO) modeling
Treatment of recovery	Not modeled	Implicit, long-term	Explicit recovery scenarios (HAZUS)
Output interpretation	Upper-bound short-term exposure	Long-term realized losses	Medium- to long-term economic disruption

complementary insights into the earliest stage of wildfire impacts.

To enable a consistent comparison, we extract and contrast the directly comparable components across studies, as summarized in Table 8. Comparability is largely limited to built-environment, property-related metrics, where our estimated building exposure of USD 29.91–55.55 billion aligns closely with UCLA Anderson's direct property loss estimates (USD 27.07 billion insured; USD 59.55 billion scaled) and LAEDC's total property damage range (USD 28.0–53.8 billion), despite substantial methodological differences. Importantly, our estimates represent upper-bound, short-term potential exposure during active fire progression, whereas UCLA Anderson and LAEDC quantify post-event realized damages and longer-term economic impacts at the county scale. Other impact domains—such as roads, POIs, and socially exposed populations—are uniquely captured in our study, while aggregate capital losses, GDP impacts, and labor income losses are reported only in regional assessments. These differences reflect distinct positions along the wildfire impact cascade, rather than inconsistencies across studies.

6. Discussion

6.1. Key findings

This study demonstrates that a fine-grained, tri-environmental assessment framework can capture wildfire exposure dynamics with

Table 5
Key quantitative estimates reported by LAEDC.

Category	Metric	Palisades	Eaton	Notes
Property damage	Total property damage (USD)	Included in total	Included in total	Combined estimate
Property damage	Total property damage range (billion USD)	~28.0–53.8	~28.0–53.8	Event-level, post-fire
Parcels affected	Parcels within burn areas (count)	10,658	9,226	Total parcels: 20,218
Parcels affected	Share of affected parcels (%)	52.7	45.6	Remaining 1.7% from other fires
Businesses affected	Businesses within fire zones (count)	1,117	746	Total businesses: 1,863
Businesses affected	Share of affected businesses (%)	60	40	—
Employment exposure	Jobs located in fire zones (count)	Included in total	Included in total	Total: 9,610 jobs
Business sale	Annual sales of affected businesses (USD)	Included in total	Included in total	$1.4 billion (total)
Business interruption losses	Lost economic output (USD, 5-year)	Included in total	Included in total	County-level
Business interruption losses	Lost output range (billion USD)	~4.6–8.9	~4.6–8.9	2025–2029
Employment impact	Job-years lost (count)	Included in total	Included in total	24,990–49,110
Labor income loss	Labor income reduction (USD)	Included in total	Included in total	$1.9–3.7 billion
Tax revenue loss	Federal/state/local tax loss (USD)	Included in total	Included in total	$0.73–1.41 billion

Table 6
Key quantitative estimates reported by UCLA Anderson.

Category	Metric	Palisades	Eaton	Notes
Property damage	Total property damage (USD)	Included in total	Included in total	Combined estimate
Property damage	Total property damage range (billion USD)	~28.0–53.8	~28.0–53.8	Event-level, post-fire
Parcels affected	Parcels within burn areas (count)	10,658	9,226	Total parcels: 20,218
Parcels affected	Share of affected parcels (%)	52.7	45.6	Remaining 1.7% from other fires
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Employment exposure	Jobs located in fire zones (count)	Included in total	Included in total	Total: 9,610 jobs
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Employment impact	Job-years lost (count)	Included in total	Included in total	24,990–49,110
Labor income loss	Labor income reduction (USD)	Included in total	Included in total	$1.9–3.7 billion
Tax revenue loss	Federal/state/local tax loss (USD)	Included in total	Included in total	$0.73–1.41 billion

strong temporal and spatial heterogeneity that are not evident from event-level summaries. By integrating daily VIIRS fire detections with land cover, infrastructure, and dasymetrically mapped population data, the framework quantifies daily exposures across the natural, built, and social environments. Based on baseline estimates aggregated across January 7–12, the total potential economic exposure from the 2025 Los Angeles wildfire is approximately US$ 48.55 billion, spanning landscape, infrastructure, and community assets in the Palisades and Eaton WUI zones.

Clear differences emerge in the timing and magnitude of exposure between the two study areas. In Palisades, exposure is strongly concentrated early in the event. Natural-environment exposure peaks on January 9 (baseline: US$ 1.78 billion), while building exposure reaches its highest daily value on January 7 (baseline: US$ 15.95 billion) and remains elevated on January 8 before declining sharply thereafter. Road-related exposure in Palisades is comparatively small in absolute value but follows a similar early-event pattern, with its maximum baseline value occurring on January 7 and steadily decreasing through January 10–11. Together, these patterns indicate that the majority of Palisades' economic exposure is front-loaded within the first days of fire progression.

In contrast, Eaton exhibits a delayed and more uneven exposure trajectory. Natural-environment exposure peaks on January 9 (baseline: US$ 398 million), similar in timing to Palisades but at a substantially lower magnitude. Building exposure in Eaton follows a multi-peak (U-shaped) pattern, with a first local maximum on January 8 (baseline: US$ 7.91 billion), a pronounced trough around January 11 (baseline: US$ 0.11 billion), and a secondary intensification on January 12 (baseline: US$ 7.37 billion). Road exposure in Eaton is minimal during the early phase but increases later in the event, with its highest baseline value also occurring on January 12. These temporal contrasts suggest that Eaton's most severe built-environment impacts emerge after the initial fire expansion into denser residential areas.

The observed differences in exposure timing are broadly consistent with the meteorological regime present during January 7–12, when Southern California experienced periods of strong offshore (Santa Ana–like) winds at the outset of the event. During such wind episodes, 10-m wind speeds in coastal foothills and passes frequently exceed 10–15 m s^{-1} (≈22–34 mph), with episodic gusts above 20 m s^{-1} (≈45 mph) reported in similar historical episodes. These conditions can accelerate fire spread along exposed ridgelines and facilitate early, high-magnitude exposure in coastal WUI neighborhoods such as Palisades. As winds moderated later in the event window, fire progression into inland suburban terrain such as Eaton may have been governed more by fuel continuity, slope, and local topography, contributing to the delayed and multi-peak exposure pattern.

Table 7
Key quantitative estimates reported by our study.

Category	Metric	Palisades	Eaton	Notes
Property damage	Total property damage (USD)	Included in total	Included in total	Combined estimate
Property damage	Total property damage range (billion USD)	~28.0–53.8	~28.0–53.8	Event-level, post-fire
Parcels affected	Parcels within burn areas (count)	10,658	9,226	Total parcels: 20,218
Parcels affected	Share of affected parcels (%)	52.7	45.6	Remaining 1.7% from other fires
Businesses affected	Businesses within fire zones (count)	1,117	746	Total businesses: 1,863
Businesses affected	Share of affected businesses (%)	60	40	—
Employment exposure	Jobs located in fire zones (count)	Included in total	Included in total	Total: 9,610 jobs
Business scale	Annual sales of affected businesses (USD)	Included in total	Included in total	$1.4 billion (total)
Business interruption losses	Lost economic output (USD, 5-year)	Included in total	Included in total	County-level
Business interruption losses	Lost output range (billion USD)	~4.6–8.9	~4.6–8.9	2025–2029
Employment impact	Job-years lost (count)	Included in total	Included in total	24,990–49,110
Labor income loss	Labor income reduction (USD)	Included in total	Included in total	$1.9–3.7 billion
Tax revenue loss	Federal/ state/local tax loss (USD)	Included in total	Included in total	$0.73–1.41 billion

Social exposures follow the same spatial and temporal divide. Population exposure in Eaton peaks on 8 January, affecting 4,342 residents who are disproportionately older, with 25.7% aged 65 or above, and predominantly White at approximately 80%, presenting heightened challenges for mobility, evacuation assistance, and targeted outreach. In contrast, Palisades experiences two high-exposure days on 8 and 9 January, with 3,926 and 1,391 exposed residents, respectively. The exposed population in Palisades is more diverse, with 54% identified as non-White and a larger share of working-age residents, implying the need for multilingual communication and sustained relief efforts. Together, these patterns demonstrate that wildfire risk in metropolitan WUI contexts is highly place-specific and time-dependent. A daily tri-environmental perspective is therefore essential for identifying who is at risk, which assets are exposed, and when critical thresholds are crossed, enabling more targeted and equitable evacuation and recovery planning.

6.2. Contribution to the literature

This study makes three distinct contributions to wildfire research. First, it operationalizes the concept of a tri-environmental framework—simultaneously evaluating natural, built, and social dimensions—through a fine-grained, data-integrated approach. Previous applications of this framework in disasters like floods and heatwaves (Aquilino et al., 2021; Wang et al., 2023) have remained largely theoretical in wildfire science. By integrating daily VIIRS fire detections, high-resolution land-cover maps, infrastructure data from OpenStreetMap, and 20-meter dasymetric population grids, this study translates the tri-environmental concept into a scalable and actionable wildfire analysis tool.

While this study adopts a tri-environmental conceptual framework, we acknowledge that the natural-environment dimension remains preliminary and proxy-based in its current implementation and does not constitute a comprehensive ecological assessment. In representing the natural environment, this study deliberately adopts land-use and land-cover (LULC) as an integrated ecological proxy. A substantial body of recent literature has demonstrated that land use change and land-cover dynamics are strongly associated with key biophysical components, including vegetation structure, habitat quality, and ecosystem service values at landscape scales (Li et al., 2025; Yetein et al., 2025; Xuan et al., 2024). For example, variations in land-use categories have been shown to influence habitat quality and associated ecosystem service values in multiple landscapes, indicating that land-use patterns reflect underlying ecological conditions (Li et al., 2025; Yetein et al., 2025). From a planning and risk-assessment perspective, LULC also represents one of the few ecological indicators that can be consistently linked to economic valuation metrics used in landscape sustainability and risk analysis

Table 8
Cross-comparison of directly comparable wildfire impact estimates.

Impact domain	Metric	Our study	UCLA Anderson (2025)	LAEDC (2025)	Notes
Spatial scope	Geographic focus	Palisades & Eaton (WUI neighborhoods)	LA County (incl. Palisades & Eaton)	Burn-area parcels (Palisades & Eaton)	Different spatial aggregation
Temporal scope	Timing	During fire progression (Jan 7–12)	Post-event	Post-event	Daily vs event-level
Natural environment	Economic exposure	3.49–9.31	—	—	Not monetized in UCLA/ LAEDC
Built environment	Buildings / property loss	29.91–55.55	27.07 (insured) / 59.55 (scaled)	28.0–53.8	Our = exposure; others = damage
Built environment	Roads / infrastructure loss	0.02–0.05	—	—	Not separately reported
Built environment	POIs affected (count)	4,508	—	—	POIs not reported elsewhere
Economic system	Total property & capital loss	—	76–131	—	County-wide, long-term
Economic output	GDP loss	—	4.6 (2025)	—	Annual flow
Labor market	Wage / labor income loss	—	0.297	—	Annual
Social exposure	Population exposed	15,861	—	—	Not estimated in UCLA/ LAEDC

(Dem et al., 2025). While variables such as species occurrence or hydrological processes offer richer ecological detail, they are difficult to monitor comprehensively and even harder to translate into comparable monetary metrics. By contrast, land-use information supports both ecological interpretation and economic quantification, aligning the natural-environment dimension of the tri-environmental framework with the practical information needs of planners and emergency managers.

Second, it bridges the gap between high-frequency environmental monitoring and dynamic social vulnerability assessment. By linking daily fire perimeters to disaggregated population surfaces, the study reveals how fire progression affects specific demographic groups—such as children, seniors, or low-income residents—within a census tract. This enables equity-sensitive planning for evacuations, sheltering, and communication strategies, and directly responds to recent calls for disaster research that centers intra-urban diversity and localized needs (Masri et al., 2021).

Third, the study contributes a replicable methodology for wildfire impact modeling that combines daily thermal satellite imagery with KDE-based perimeter estimation, infrastructure overlays, and POI-specific exposure metrics. This integrated pipeline allows for the day-by-day tracking of fire dynamics and their cascading impacts on ecosystems, buildings, and human services. Unlike seasonal or post hoc damage assessments, this workflow provides emergency managers with temporally precise and spatially explicit insights to support real-time decision-making and post-disaster resource allocation.

6.3. Policy implications

The results highlight several ways Los Angeles and other wildfire-prone cities can translate a tri-environmental perspective into practice. First, emergency protocols must be timed and tailored to local demographics. Eaton's older population requires evacuation support and medical continuity, while Palisades' cultural diversity demands multilingual alerts and targeted outreach. Land-use policies should reflect localized risk: in Eaton, where business corridors face high losses, fire-resistant standards must apply to shops and mixed-use buildings—not just homes. In Palisades, where public spaces lie within the burn zone, fire-hardening measures like ember-resistant roofs, redundant water, and fire-safe landscaping are essential. Real-time data integration is also key: daily VIIRS detections and short-term spread models should feed into municipal dashboards to enable proactive deployment of personnel, air support, and traffic control.

Secondly, the study highlights the importance of integrating high-resolution impact data into long-term resilience planning. Instead of relying solely on post-fire damage surveys or coarse county-level statistics, municipalities should adopt spatially disaggregated assessment tools like the tri-environmental framework used here. This enables planners to evaluate not only where fire has occurred, but also which infrastructure and communities are repeatedly at risk, allowing for more adaptive zoning, strategic retrofits, and targeted investment in both physical and social resilience.

Finally, comparative evidence from other wildfire-prone regions underscores the generalizability of this tri-environmental framework while also revealing context-specific vulnerabilities. In Australia, empirical studies show that house loss is strongly determined by nearby fuels, slope, and local housing density, linking the natural environment and the built environment while highlighting the importance of tailored measures in the social environment (Price & Bradstock, 2013). In Mediterranean Europe, cross-border simulations integrate harmonized fuels, wind, and slope with settlement footprints to quantify exposure and evaluate cost-effective WUI fuel removals, illustrating how the natural environment and the built environment can be optimized under varied governance systems (Salis et al., 2023). In Northern Europe, sparse settlements reveal that U.S.-style density thresholds underestimate exposure; instead, ember-transport distances and dedicated fuel maps are more relevant, underscoring the role of the natural environment and the built environment in boreal contexts and the need for adaptive strategies in the social environment (Medina et al., 2025). In North America, California case studies reveal that housing arrangement, edge proximity, and community demographics interact with wind-driven fire weather to shape structure survival, while nationwide parcel-level hazard models demonstrate how building-level WUI mapping improves prediction of losses, showing how the social environment, natural environment, and built environment must be considered together (Kearns et al., 2022). A Pacific Northwest mapping effort further shows that fine-scale building footprints and address data yield more accurate WUI baselines than census units, enhancing the built environment component of risk assessment (Wei et al., 2023). Beyond these core regions, Chile has pioneered machine-learning approaches that calibrate WUI mapping to modeled fire probabilities, bridging the natural environment with settlement density in the built environment (Miranda et al., 2020), while studies in Israel/West Bank integrate observed fire activity with building–fuel combinations to delineate hotspots, offering lessons on governance and preparedness in the social environment (Tikotzki et al., 2023). Together, these cases confirm that wildfire risk consistently emerges from the interaction of the social environment, the natural environment, and the built environment, even though the dominant drivers vary across contexts—from wind-driven fires in California and the Mediterranean to fuel-continuity challenges in Australia and sparse-settlement dynamics in Scandinavia—demonstrating both the generalizability and adaptability of the tri-environmental framework across diverse regions.

6.4. Limitations

This study has several limitations that should be acknowledged. A key limitation of this study is that the reported economic values represent potential economic exposure rather than observed or realized damages. In our framework, exposure is defined as the maximum replacement value of assets intersecting with daily fire perimeters under the assumption of complete destruction. This proxy inevitably overlooks mitigating factors such as fire suppression effectiveness, construction type and resilience, and the presence of defensible space, and it does not differentiate between exposures occurring under mild fire conditions versus those under extreme wind and low humidity. Consequently, the estimates should be interpreted as upper-bound approximations of wildfire-related economic risk rather than precise accounts of actual losses. Future research should seek to refine these estimates by incorporating burn severity indices such as dNBR or RdNBR, meteorological and fuel variables including wind speed, humidity, and fuel moisture (Abatzoglou et al., 2023), and fire behavior models that simulate intensity and spread dynamics (Chen et al., 2022). These additions would enable stratification of exposure according to fire severity and the estimation of partial rather than total losses. Moreover, combining the exposure framework with post-fire field surveys or insurance and claims data (Monge et al., 2023) could help validate modeled outcomes against realized damages. By linking potential exposure with observed impacts, future studies will improve both the accuracy and policy relevance of tri-environmental wildfire assessments.

Second, the representation of the natural environment in this study is simplified and relies primarily on land-use/land-cover (LULC) as an integrated proxy. This study does not aim to deliver a comprehensive ecological impact assessment at the level of individual species, soil processes, or hydrological dynamics. While post-fire hydrological changes, including altered runoff, erosion, and soil moisture dynamics, represent important and often direct ecological impacts (Badia et al., 2019; Kim et al., 2025), incorporating these processes requires high-resolution hydrological and soil data and process-based modeling frameworks that are beyond the scope of the present regional-scale, planning-oriented analysis. Instead, consistent with previous wildfire damage and vulnerability studies that rely on satellite-based land-cover

information (Seydi et al., 2021), this study focuses on indicators that can be systematically observed and linked to valuation-oriented exposure assessment. Consequently, our analysis emphasizes built and social dimensions while recognizing the need for future work to integrate ecological and hydrological process models to strengthen environmental representation. Future research should expand this dimension by incorporating dynamic indicators that capture ecological impacts more comprehensively. Remotely sensed vegetation indices such as NDVI have long been established for assessing burn severity and post-fire vegetation recovery, and recent studies have applied these methods to track multi-year recovery trajectories in European forests (Lopes et al., 2024; Wassner et al., 2025). In addition, new work highlights systematic fire effects on plant–soil water relations, showing how fire alters soil moisture and vegetation water use dynamics in ways not captured by static land-cover proxies (Baur et al., 2024). Taken together, these advances demonstrate that while LULC remains a robust and operational proxy, integrating vegetation indices, soil moisture dynamics, and ecological resilience measures will provide a more complete representation of the natural environment in future wildfire impact assessments.

Third, although this study incorporates scenario-based uncertainty ranges to partially address valuation variability, more advanced approaches could further improve uncertainty characterization in future work. Recent studies suggest that spatially explicit hedonic pricing models, leveraging high-resolution parcel data and neighborhood attributes, offer a robust way to capture heterogeneity in land and property values across urban areas (e.g., Athukorala et al., 2019). In addition, probabilistic uncertainty propagation frameworks, such as Monte Carlo–based loss modeling widely used in contemporary disaster risk assessments, can jointly account for uncertainty across multiple valuation and exposure parameters rather than relying on fixed percentage bounds (e.g., Lautenberger, 2017). Integrating these approaches would allow future analyses to generate statistically grounded uncertainty intervals and improve comparability with post-event economic loss assessments.

In addition, the study is geographically limited to two urban WUI areas in Los Angeles. While Eaton and Palisades provide valuable contrast in topography and demographics, the findings may not fully represent wildfire dynamics or vulnerabilities in other settings. Broader generalization would benefit from applying the framework to a wider range of locations, including rural or lower-income urban peripheries with different built and social conditions. Future studies could also enhance the model by integrating higher-resolution fire detection, mobility data, and functional weighting of critical building. Despite these limitations, the tri-environmental framework offers a scalable and replicable structure for advancing urban wildfire impact analysis and informing more targeted and inclusive adaptation strategies.

7. Conclusion

This study presents a tri-environmental framework that integrates daily satellite fire detections with natural, built, and social environment data to assess wildfire exposures in Los Angeles's Eaton and Palisades WUI zones. By tracking day-to-day flame-front shifts, quantifying exposures to land, roads, buildings, and community services, and identifying which age, income, and racial groups are most at risk, the framework demonstrates that wildfire is simultaneously an ecological, infrastructural, and social crisis whose impacts vary block by block. These findings highlight that city-wide, generic strategies are insufficient; instead, interventions—such as evacuation support, zoning updates, fuel management, and equitable recovery funds—must be matched to the specific demographic, infrastructural, and ecological profiles of each neighborhood. Beyond these research contributions, the framework provides clear pathways for operational integration. For real-time alerting, it can be linked with existing systems such as Alert LA County and the Ready, Set, Go! program (Los Angeles County Fire Department, 2021), embedding satellite detections and CAL FIRE's 2025 Fire Hazard Severity Zone maps into municipal dashboards to trigger evacuation alerts and resource deployment (CAL FIRE, 2025). For predictive adaptation planning, it complements mandates such as California Public Resources Code §4291 on defensible space (California Legislature, 2020) and Los Angeles County's Countywide Community Wildfire Protection Plan (Los Angeles County, 2019), by highlighting zones where fuel reduction, fire-resistant retrofits, or targeted subsidies are most urgent—particularly in elderly or low-income neighborhoods with limited adaptive capacity. For municipal resilience pathways, the framework can be embedded into the City of Los Angeles's forthcoming Community Wildfire Protection Plan (City of Los Angeles, 2023), guiding community-driven identification of evacuation corridors, vulnerable facilities, and preparedness programs, and linking these to zoning codes, insurance incentives, and long-term land-use strategies. By demonstrating both scientific rigor and policy applicability, this framework bridges high-resolution wildfire analytics with actionable governance tools. As climate change and urban expansion intensify wildfire threats, such integrated, context-specific approaches will be essential for building communities that are not only fire-adapted but also socially just.

CRediT authorship contribution statement

Zongrong Li: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haiyang Li:** Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Yifan Yang:** Writing – review & editing, Validation, Methodology, Investigation, Data curation, Conceptualization. **Siqin Wang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Yingxin Zhu:** Writing – original draft, Software, Investigation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

Appendix A

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Table A1.1
Daily Baseline and Uncertainty-Bounded Estimates of Potential Wildfire Exposure to Natural and Built Environments in Los Angeles (January 7–12, 2025).

District	Category	Unit	Scenario	Jan.7th	Jan.8th	Jan.9th	Jan.10th	Jan.11th	Jan.12th
Palisades	Natural Environment	US Dollar	Low (−40%)	936,834,234.60	744,522,639	1,068,254,773.20	143,462,827.20	2,511,588.73	6,077,507.90
Palisades	Natural Environment	US Dollar	Baseline	1,561,390,391	1,240,871,065	1,780,424,622	239,104,712	4,185,981.21	10,129,179.83

(continued on next page)

Table A1.1 (*continued*)

District	Category	Unit	Scenario	Jan.7th	Jan.8th	Jan.9th	Jan.10th	Jan.11th	Jan.12th
Palisades	Natural Environment	US Dollar	High (+60%)	2,498,224,625.60	1,985,393,704	2,848,679,395.20	382,567,539.20	6,697,569.94	16,206,687.73
Palisades	Built Environment: Buildings	US Dollar	Low (−30%)	11,163,591,551	4,615,750,204.80	28,676,677.43	14,338,338.71	17,922,923.40	3,291,843,597
Palisades	Built Environment: Buildings	US Dollar	Baseline	15,947,987,930	6,593,928,864	40,966,682.04	20,483,341.02	25,604,176.28	4,702,633,710
Palisades	Built Environment: Buildings	US Dollar	High (+30%)	20,732,384,309	8,572,107,523.20	53,256,686.65	26,628,343.33	33,285,429.16	6,113,423,823
Palisades	Built Environment: Roads	US Dollar	Low (−40%)	534,502.91	141,712.67	60,029.57	27,687.57	27,153.72	57,789.42
Palisades	Built Environment: Roads	US Dollar	Baseline	890,838.19	236,187.78	100,049.28	46,145.95	45,256.21	96,315.71
Palisades	Built Environment: Roads	US Dollar	High (+40%)	1,247,173.46	330,662.89	140,069.00	64,604.34	63,358.69	134,842.00
Palisades	Built Environment: Points of Interest	Number	Baseline	220	1397	1045	14	6	8
Eaton	Natural Environment	US Dollar	Low (−40%)	—	216,611,946.90	239,042,706.36	7,036,073.72	8,397,576.64	117,783,880.26
Eaton	Natural Environment	US Dollar	Baseline	—	361,019,911.50	398,404,510.60	11,726,789.54	13,995,961.07	196,306,467.10
Eaton	Natural Environment	US Dollar	High (+60%)	—	577,631,858.40	637,447,216.96	18,762,863.26	22,393,537.71	314,090,347.36
Eaton	Built Environment: Buildings	US Dollar	Low (−30%)	—	5,537,197,246	2,495,919.43	4,367,858.99	74,877,582.78	5,158,441,472.80
Eaton	Built Environment: Buildings	US Dollar	Baseline	—	7,910,281,780	3,565,599.18	6,239,798.56	106,967,975.40	7,369,202,104
Eaton	Built Environment: Buildings	US Dollar	High (+30%)	—	10,283,366,314	4,635,278.93	8,111,738.13	139,058,368.02	9,579,962,735.20
Eaton	Built Environment: Roads	US Dollar	Low (−40%)	—	535,046.70	37,624.62	28,467.07	18,603.74	641,078.83
Eaton	Built Environment: Roads	US Dollar	Baseline	—	891,744.50	62,707.70	47,445.12	31,006.24	1,068,464.71
Eaton	Built Environment: Roads	US Dollar	High (+40%)	—	1,248,442.30	87,790.78	66,423.17	43,408.73	1,495,850.59
Eaton	Built Environment: Points of Interest	Number	Baseline	—	901	915	2	0	6

Table A1.2
Composition of Fire-Affected Land Cover Types in the Natural Environment.

Category: Natural Environment	Palisades	Eaton
Residential	92.73%	98.32%
Industrial	4.43%	0.26%
Grass	2.69%	—
Others	0.15%	1.42%

Table A1.3
Composition of Fire-Affected Roads in the Built Environment.

Category: Roads	Palisades	Eaton
Pedestrian Paths	42.76%	10.21%
Primary Roads	28.45%	30.06%
Residential & Service Roads	6.21%	78.91%
Tracks	2.47%	9.15%
Unclassified Roads	0.01%	0.00%

Table A1.4
Composition of Fire-Affected Points of interest in the Built Environment.

Category: Points of Interest	Palisades	Eaton
Business and Prof. Services	24.08%	34.74%
Community and Government	5.23%	13.55%
Dining and Drinking	9.94%	9.41%
Landmarks and Outdoors	6.85%	14.99%
Retail	11.46%	13.63%
Unknown	6.28%	7.05%
Others	12.76%	13.64%

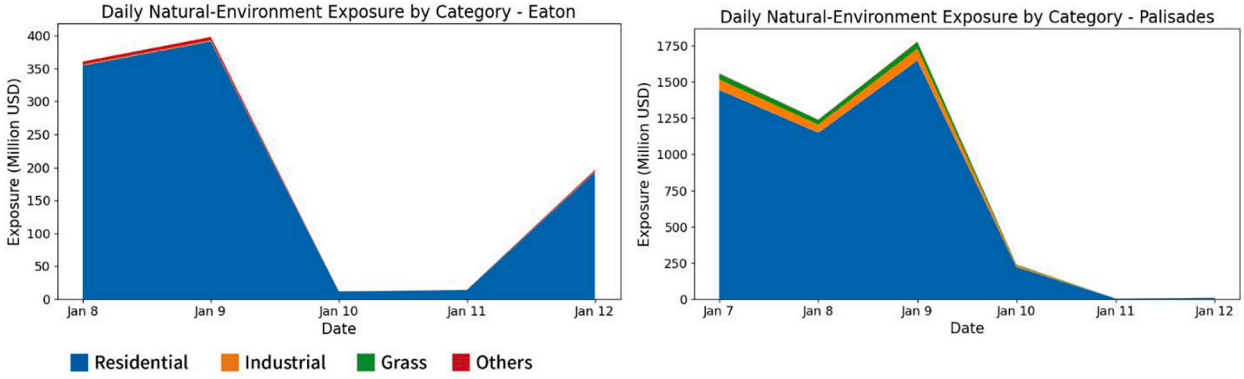


Fig. A1. Temporal Changes and Composition of Wildfire-Affected Land Use (Stacked Area Charts)

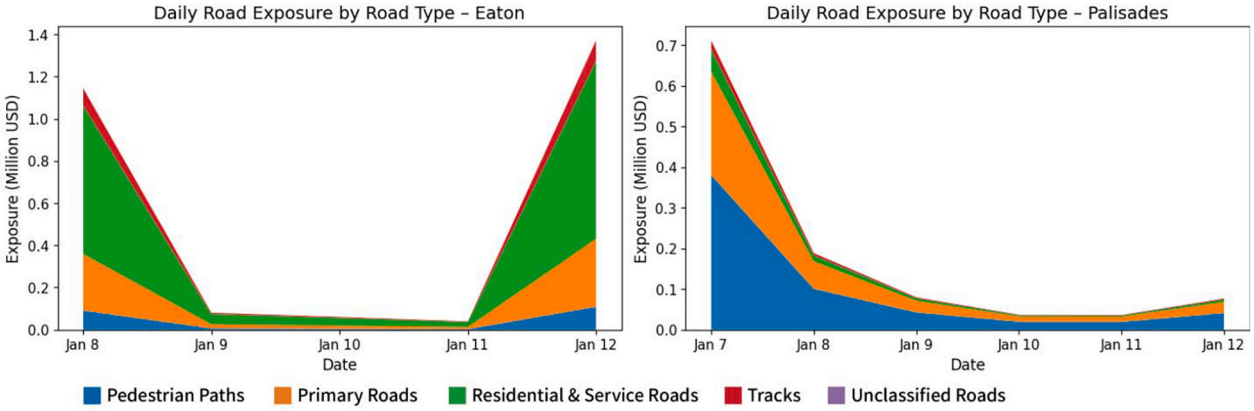


Fig. A2. Temporal Changes and Composition of Wildfire-Affected Roads (Stacked Area Charts)

Data availability

Data will be made available on request.

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